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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations
December 2025

EE2101-2 – ANALOG SIGNAL PROCESSING
(For EEE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. A MOSFET operating in the triode region behaves like a
 - A) Amplifier
 - B) Switch Off
 - C) Voltage controlled resistor
 - D) Current controlled resistor
2. What is the purpose of adding a resistor in the source terminal in a MOSFET amplifier?
 - A) Stabilize bias point
 - B) Decrease gain
 - C) Increase gain
 - D) Increase input resistance
3. The MOSFET acts as a linear amplifier when it operates in which region?
 - A) Cutoff
 - B) Saturation
 - C) Triode
 - D) Breakdown
4. What does MOSFET stand for?
 - A) Metal Oxide Semiconductor Field Emission Transistor
 - B) Metal Oxide Semiconductor Field-Effect Transistor
 - C) Metal Oxide Semiconductor Fast-Effect Transistor
 - D) Metal Oxide Semiconductor Floating-Effect Transistor
5. In the MOSFET's cutoff region, the device acts as a
 - A) Conductor
 - B) Switch (ON)
 - C) Amplifier
 - D) Switch (OFF)
6. For a MOSFET, the input signal is typically applied to which terminal?
 - A) Source
 - B) Drain
 - C) Gate
 - D) Body
7. What is the primary function of the oxide layer in a MOSFET?
 - A) Conduct current
 - B) Isolate the gate terminal
 - C) Control gate current
 - D) Provide feedback
8. The saturation region of a MOSFET is characterized by:
 - A) Constant drain current (I_D)
 - B) Exponential increase in drain current (I_D)
 - C) Zero drain current (I_D)
 - D) Non-linear relationship between I_D and V_{GS}
9. What is the primary function of a common source MOS amplifier?
 - A) Amplify signals with a high input impedance
 - B) Amplify signals with a low output impedance
 - C) Act as a current source
 - D) Provide voltage buffering
10. The common drain amplifier is also known as
 - A) Current amplifier
 - B) Voltage amplifier
 - C) Source follower
 - D) Resistor amplifier

11. The phase difference between the input and output of a Common Source (CS) amplifier is
 - A) 0°
 - B) 45°
 - C) 90°
 - D) 180°
12. The primary property of negative feedback in amplifiers is to
 - A) Increase distortion
 - B) Improve stability
 - C) Decrease bandwidth
 - D) Increase power consumption
13. In a common source amplifier, increasing the load resistance will
 - A) Decrease the gain
 - B) Increase the gain
 - C) Have no effect on the gain
 - D) Decrease the output impedance
14. For a MOSFET amplifier, the higher-frequency cutoff is mainly determined by
 - A) Internal capacitances
 - B) The transistor's threshold voltage
 - C) Biasing resistors
 - D) The power supply voltage
15. For a common drain amplifier, the voltage gain is typically
 - A) Greater than one
 - B) Less than one
 - C) Zero
 - D) Infinite
16. The Miller effect increases
 - A) Effective resistance
 - B) Effective output capacitance
 - C) Effective input capacitance
 - D) Effective voltage gain
17. What is the primary characteristic of an ideal operational amplifier?
 - A) Infinite input impedance and zero output impedance
 - B) Zero input impedance and infinite output impedance
 - C) Finite input impedance and finite output impedance
 - D) Zero gain and infinite bandwidth
18. Ideal value of CMRR is
 - A) Zero
 - B) Infinity
 - C) 1
 - D) 1000
19. An inverting amplifier has a phase shift of
 - A) 0 degrees
 - B) 90 degrees
 - C) 180 degrees
 - D) 270 degrees
20. A Schmitt trigger circuit is primarily used for
 - A) Converting an analog signal to a digital signal
 - B) Generating precise frequency signals
 - C) Filtering high-frequency noise
 - D) Providing a stable reference voltage

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) With internal structure elaborate working principle of MOSFET with v_{GS} greater than V_t and v_{DS} very small value.	6	L2	1	3
	b) Consider an NMOS transistor fabricated in a $0.7\ \mu\text{m}$ process with $W = 14\ \mu\text{m}$, $t_{ox} = 10\text{nm}$, $\mu_n = 500\ \frac{\text{cm}^2}{\text{Vs}}$. Find V_{ov} required to operate MOSFET in saturation with $I_D = 0.1\text{mA}$.	5	L3	1	3
	c) With a neat circuit diagram explain biasing of the MOSFET with gate to drain resistance.	5	L2	2	3
2.	a) Deduce an expression for drain current i_D of n- channel MOSFET in saturation and triode region.	6	L2	1	3
	b) With a neat circuit diagram explain biasing of the MOSFET with source resistance.	5	L2	2	3
	c) Deduce T equivalent model of CS amplifier from pi model.	5	L2	2	2

3. a) Deduce the small signal model of MOSFET and elaborate significance of transconductance.
- b) With a neat circuit diagram explain current source biasing of the MOSFET.
- c) MOSFET circuit shown in figure 3c. has V_{DD} 3.3V, $V_D=0.1V$, $V_t=1V$, $k_n=1mA/V^2$. Find value of R_D .

6	L2	1	3
5	L1	2	2

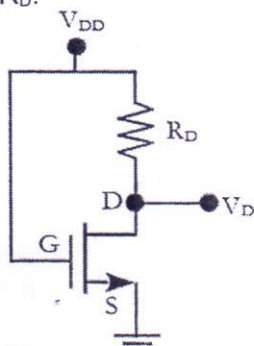


Figure 3c.

Unit – II

4. a) Describe amplifier characteristics and specific application of Common Drain amplifier.
- b) With necessary equivalent circuit describe unity gain frequency of CS amplifier.
- c) A CS amplifier utilizes a MOSFET biased at $I_D = 0.5$ mA with $V_{OV} = 0.25$ V and $R_D = 25k\Omega$. The amplifier is fed with a source having $R_{sig} = 100k\Omega$, and a $25k\Omega$ load is connected to the output. Find R_{in} , A_{vo} , R_o , A_v and G_v .
5. a) Describe different types of internal capacitances in MOSFET.
- b) With necessary equivalent circuit describe low frequency response of a CS amplifier.
- c) Describe the operation of a CMOS logic inverter.
6. a) With necessary equivalent circuit describe high frequency response of a CS amplifier.
- b) Describe shunt-series feedback amplifier using block diagram.
- c) A CG amplifier is designed with $g_m=1mA/V$, $R_D=20k\Omega$, $R_L=20k\Omega$, $R_{sig}=100\Omega$. Find
- R_{in}
 - A_v
 - R_o
 - G_v

5	L3	2	2
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5	L2	3	3
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5	L2	4	3
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6	L*3	3	3
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5	L2	4	3
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6	L2	4	3
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5	L2	3	3
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6	L2	3	3
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6	L1	4	3
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4	L3	3	3
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Unit – III

7. a) With necessary equivalent circuit elaborate common mode gain and CMRR.
- b) With neat diagram explain the internal architecture of OPAMP.
8. a) Design a capacitor coupled non inverting amplifier with a load resistor R_L of $2.2k\Omega$, gain 50, input voltage 20mV and lower cut-off frequency of 150Hz. Assume 741 opamp.
- b) Explain the operation of inverting Schmitt trigger circuit with neat circuit diagram and waveforms
- c) Discuss steps to design first order low pass filter.

8	L2	5	3
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8	L2	5	3
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6	L3	5	3
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5	L2	6	3
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5	L2	5	3
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BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
